# SVAC-Concurrency: A Step-Verification Benchmark for Large Language Model Reasoning on Operating System Concurrency Algorithms

Gagan Tyagi[1], Aneesh Kumar[1]
[1]Department of Computer Science and Engineering,
JECRC University, Jaipur, Rajasthan, India
Email: {gagan.tyagi, aneesh.kumar}@jecrcuniversity.edu.in

***Abstract*—While Large Language Models (LLMs) demonstrate remarkable competence in natural language understanding and high-level source code synthesis, their ability to faithfully track deterministic, multi-step algorithmic state transitions remains fundamentally unverified. Concurrency control and resource allocation algorithms in Operating Systems (OS) represent a rigorous, safety-critical testbed for mechanistic reasoning, demanding strict invariant preservation, vector/matrix state updates, and topological cycle detection.**

**In this paper, we present SVAC-Concurrency, an open-source, reproducible evaluation benchmark comprising 125 parameterized problem instances (generating 375 evaluation traces per model across 3 prompting regimes) spanning five canonical OS algorithms: Wait-For Graph (WFG) deadlock detection, Banker's safety algorithm, Buddy System memory allocation, counting Semaphores, and Mutex lock simulations.**

**Rather than relying solely on coarse outcome correctness (Verdict Accuracy), SVAC-Concurrency introduces a deterministic ground-truth verification framework to evaluate execution traces across four diagnostic dimensions: Step Accuracy (SA), First Error Position (FEP), Error Cascade Rate (ECR), and Equivalence-Aware Invariant Validity ($SA_{valid}$). We conduct an extensive cross-architectural investigation evaluating Google Gemini-3.5-Flash-Lite, OpenAI GPT-OSS-20B, NVIDIA Nemotron-3-Super-120B, and Liquid AI LFM-2.6B across Zero-Shot, Few-Shot, and Chain-of-Thought (CoT) prompting strategies.**

**Our empirical findings reveal:**

1) **Procedural Fidelity vs. Outcome Illusion: On non-saturating state-tracking tasks (WFG, Banker's, Buddy), Gemini-3.5-Flash-Lite achieves a mean Verdict Accuracy of 59.7% on the same tasks where its mean strict Step Accuracy drops to 36.99% SA (a 22.7 pp gap), demonstrating widespread "right verdict for flawed reasons" behavior.**
2) **Prompt Conventions vs. Invariant Validity in Graph Reasoning: Under strict positional matching, WFG cycle detection exhibits an apparent collapse from 85.4% SA in Few-Shot to 7.5% in CoT (and 4.4% in Zero-Shot). Crucially, equivalence-aware invariant verification reveals that CoT maintains 93.2% valid step accuracy and 93.3% verdict accuracy (comparable to Few-Shot's 100.0% valid SA). The collapse in strict positional SA is driven by prompt/solver specification conventions for initial visited node tracking rather than algorithmic breakdown.**
3) **Failure Horizon Dichotomy: WFG strict positional errors initiate at Step 1 in 100% of Zero-Shot and CoT cases due to initial set convention mismatches, whereas Buddy allocation exhibits gradual multi-step arithmetic fragmentation errors (mean FEP $= 6.55$–$7.57$ steps).**
4) **Equivalence vs. Strict Positional Bias: Invariant-based scoring demonstrates that strict positional matching severely underestimates model competence: valid step accuracy increases to 74.3% in Banker's ($\Delta = +32.8$ pp) and 95.6% in WFG ($\Delta = +64.3$ pp).**
5) **Network and Format Fragility in Open Architectures: Open-weight API runs suffered heavily from upstream rate-limiting (e.g., 262/375 empty calls for Nemotron-120B). Among non-empty responses received, parsing success reached 73.5% for Nemotron-120B, 88.9% for GPT-OSS-20B, and 97.4% for LFM-2.6B.**

**The complete benchmark suite, deterministic reference solvers, prompt templates, and evaluation pipelines are publicly available at: https://github.com/Bethegnt/SVAC-Concurrency.**

***Index Terms*—Large Language Models, Algorithmic Reasoning, Operating Systems, Concurrency Control, Deadlock Detection, Banker's Algorithm, Step-level Verification, Benchmark.**

## I. Introduction

Large Language Models (LLMs) are increasingly integrated into modern software engineering pipelines, automated code review tools, and operating systems pedagogy [1], [2]. Despite remarkable empirical gains on functional coding benchmarks such as HumanEval [3], MBPP [4], and APPS [5], conventional benchmarks evaluate programs almost exclusively via black-box input-output test cases. They fail to assess whether an auto-regressive model maintains a coherent, mechanically faithful simulation of intermediate program states during execution [6]–[8].

In operating systems and systems programming, procedural correctness is non-negotiable. Concurrency control algorithms—such as deadlock detection, resource allocation safety, and dynamic physical memory partitioning—are governed by rigorous theoretical properties:

1) **Strict Discrete State Tracking:** Every transition must update resource availability vectors, process allocation matrices, or block allocation trees without state drift or phantom resource generation.
2) **Topological and Relational Invariants:** Identifying deadlocks requires systematic traversal of directed Wait-For Graphs to detect closed cycles without hallucinating non-existent dependency edges.
3) **Catastrophic Error Cascades:** A single state corruption at step $t$ invalidates all subsequent state assertions at $t+k$, converting downstream execution into an unfaithful hallucination.

When evaluating LLMs on such procedural tasks, standard outcome-based metrics (e.g., binary classification of whether a system is deadlocked or safe) introduce massive evaluation noise. On balanced datasets, an LLM can easily guess the correct binary verdict through statistical heuristics or shallow token correlations while executing an entirely invalid sequence of intermediate states.

To resolve this evaluation gap, we introduce **SVAC-Concurrency** (*Step-Verification of Algorithmic Concurrency*), a comprehensive, parameter-controlled benchmark designed to evaluate intermediate LLM reasoning traces across five foundational OS concurrency algorithms. Each instance is paired with an exact ground-truth execution trace computed by deterministic reference solvers, unit-tested against textbook worked examples [9] and, for WFG and Banker's algorithms (60 of 125 instances), validated via independent brute-force differential testing.

*Research Questions*

- **RQ1** (*Procedural Fidelity vs. Outcome Correctness*): Do LLMs produce faithful intermediate state transitions, or do high verdict accuracies mask severe intermediate state-tracking failures on balanced decision distributions?
- **RQ2** (*Complexity Scaling & Failure Horizons*): How do increasing state-space bounds affect step accuracy, and at what execution horizon (FEP) do cascading errors initiate across different algorithmic structures?
- **RQ3** (*Prompting Dynamics & Serialization Artifacts*): Does step-by-step verbalization (CoT) assist or destabilize discrete graph-traversal and matrix-tracking algorithms, and to what extent does few-shot prompting act as format serialization priming?
- **RQ4** (*Equivalence & Architectural Robustness*): To what extent does strict positional scoring underestimate valid alternative execution paths, and how do open-weight models adhere to strict intermediate state schemas compared to instruction-tuned frontier APIs?

*Primary Contributions*

- **Comprehensive Benchmark Suite:** 125 unique instances across 5 core OS algorithms parameterized across 5 distinct complexity tiers ($N$=375 evaluations per model across 3 prompt strategies).
- **Multi-Dimensional Metric Taxonomy:** Formalization of SA, FEP, ECR, and Equivalence-Aware Invariant Validity to capture reasoning dynamics beyond binary verdicts.
- **Cross-Architectural Empirical Investigation:** Systematic comparison of four distinct model families: Gemini-3.5-Flash-Lite, GPT-OSS-20B, Nemotron-3-Super-120B, and Liquid AI LFM-2.6B.
- **Deconstruction of the CoT Inversion Effect:** Discovery and formal characterization showing that the apparent collapse of CoT in graph cycle detection (85.4% Few-Shot vs. 7.5% CoT strict SA) is primarily driven by serialization tie-breaking divergence, while invariant-aware step validity remains robust at 93.2% with 93.3% verdict accuracy.
- **Equivalence-Aware Invariant Evaluation:** Demonstration that rigid positional matching under-reports valid algorithmic execution by 32.8% in Banker's and 64.3% in WFG.

## II. Related Work

### A. Benchmarking Code Generation and Execution Reasoning

Early LLM code evaluation benchmarks such as HumanEval [3], MBPP [4], and APPS [5] established input-output testing as the standard paradigm for evaluating program synthesis. Recent efforts have shifted toward evaluating *execution reasoning*. CRUXEval [10] evaluated code execution reasoning by testing input/output prediction across short Python snippets. REval [11] and CodeMind [12] introduced state-level execution tracing for general imperative code. While these benchmarks test arbitrary code snippets, they do not systematically probe canonical discrete algorithms where formal invariants, resource constraints, and topological cycles dictate correctness. SVAC-Concurrency provides precisely this specialized, parameter-controlled algorithmic framework.

### B. Mechanistic State Tracking in Transformers

A growing body of literature investigates whether transformer architectures maintain coherent internal world models during sequential tasks. Toshniwal et al. [13] demonstrated that auto-regressive language models track board state representations in chess games despite receiving only move sequences. Strobl et al. [14] surveyed the theoretical boundaries of what formal languages and state machines transformer architectures can express. Wang et al. [15] investigated graph traversal algorithms in natural language. In the systems domain, the precursor SVAC technical report [16] evaluated step-level reasoning in virtual memory page replacement policies (FIFO, LRU, Optimal, Clock). Additionally, the CONCUR benchmark by Huang et al. [17] evaluated LLMs on *synthesizing* concurrent synchronization primitives. Our work explicitly contrasts with CONCUR: rather than evaluating concurrent code synthesis, we evaluate *execution-trace verification*, isolating whether models accurately track concurrent state transitions and deadlock invariants.

### C. Prompting Dynamics and CoT Unfaithfulness

Chain-of-Thought (CoT) prompting [18], [19] is widely documented to boost arithmetic and multi-hop reasoning. However, recent empirical studies reveal significant limits and pathology in verbalized reasoning. Lanham et al. [20] and Turpin et al. [21] demonstrated that CoT explanations are frequently unfaithful, serving as post-hoc rationalizations rather than causal computation drivers. Shi et al. [22] demonstrated that LLMs are easily distracted by irrelevant reasoning context, while Liu et al. [23] observed degradation in multi-step planning tasks when natural language narration exceeds formal state-space capacity. In graph-theoretic tasks, Agrawal et al. [24] investigated the structural limitations of language

models on graph reasoning. Our work grounds these phenomena in operating systems concurrency, providing quantitative evidence of how serialization tie-breaks and verbalized narration interact with discrete state tracking.

## III. Benchmark Architecture and Methodology

### A. Problem Formulations & Ground-Truth Verification

All problem instances are deterministically synthesized across five calibrated complexity tiers (Levels 1–5). Ground-truth execution traces are produced by reference Python solvers (`solvers/`) unit-tested against textbook worked examples from Silberschatz et al. [9]. Additionally, WFG and Banker's Algorithm solvers (60 of 125 instances) were independently cross-verified via brute-force differential testing against alternative reference implementations (3-color DFS oracle for WFG; exhaustive permutation search for Banker's).

1) **Wait-For Graph (WFG) Deadlock Detection:** Given a set of processes $P = \{P_0, \ldots, P_{n-1}\}$ ($|V| \in [3, 12]$, $|E| \in [2, 15]$) and directed wait edges $E \subseteq P \times P$, where $(P_i, P_j) \in E$ indicates $P_i$ is blocked waiting for a resource held by $P_j$. The model must execute DFS, tracking the active recursion stack and visited set at each step, identifying back-edges, reporting the cycle subgraph if present, and emitting a final verdict deadlock $\in \{\text{true}, \text{false}\}$. The dataset is strictly balanced: 15 instances contain cycles and 15 are directed acyclic graphs (DAGs), balanced evenly (3 positive, 3 negative) across all 5 complexity tiers.
2) **Banker's Algorithm for Deadlock Avoidance:** Given an Available resource vector $A \in \mathbb{N}^m$, Allocation matrix $\mathbf{Alloc} \in \mathbb{N}^{n \times m}$, and Max demand matrix $\mathbf{Max} \in \mathbb{N}^{n \times m}$, parameterized by $P \in [3, 8]$ processes and $R \in [2, 4]$ resource types. The model must compute $\mathbf{Need} = \mathbf{Max} - \mathbf{Alloc}$, iteratively identify an executable candidate process $P_i$ such that $\mathbf{Need}_i \leq \text{Work}$, simulate resource reclamation $\text{Work} \leftarrow \text{Work} + \mathbf{Alloc}_i$, update the Finish boolean array, and determine whether a complete safe sequence exists (safe $\in \{\text{true}, \text{false}\}$). Strictly balanced: 15 safe and 15 unsafe instances (3 per complexity tier).
3) **Buddy System Memory Allocation:** Given total memory size $M \in [32, 512]$ units and a sequential stream of 1 to 20 allocation and deallocation operations. The model must track recursive block splits, power-of-two alignments, and buddy coalescing on free operations, returning the exact block partition state at each step.
4) **Counting Semaphore Simulation:** Given initial counter $S_0 \in [1, 3]$ and an interleaved execution stream of 4 to 14 concurrent wait($S$) and signal($S$) calls from multiple threads. The model must track the integer value of $S$ and the FIFO blocked thread queue at every transition.
5) **Mutex Lock Simulation:** Evaluates mutual exclusion lock acquisition, priority ownership, contention blocking, and release transitions across 2 to 6 concurrent threads over streams of 3 to 14 operations.

### B. Mathematical Evaluation Framework

Let an evaluation instance $I$ have ground-truth trace $\mathbf{Y}^* = (y_1^*, y_2^*, \ldots, y_{T^*}^*)$ of length $T^*$, and model-generated trace $\hat{\mathbf{Y}} = (\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_m)$ of length $m$.

*1) Formal Step Equality Specification:* A step $\hat{y}_t$ matches ground truth $y_t^*$ if and only if all task-specific state attributes match under canonical normalization:

**WFG Deadlock:**

$$\begin{aligned} \hat{y}_t = y_t^* \iff & \big(\hat{y}_t.\text{action} = y_t^*.\text{action}\big) \\ & \wedge \big(\hat{y}_t.\text{node} = y_t^*.\text{node}\big) \\ & \wedge \big(\text{sort}(\hat{y}_t.\text{visited}) = \text{sort}(y_t^*.\text{visited})\big) \\ & \wedge \big(\hat{y}_t.\text{stack} = y_t^*.\text{stack}\big) \end{aligned} \tag{1}$$

Visited nodes are compared as canonical sorted sets (order-insensitive), while the recursion stack is compared as an ordered sequence (order-sensitive).

**Banker's Algorithm:**

$$\begin{aligned} \hat{y}_t = y_t^* \iff & \big(\hat{y}_t.\text{candidate} = y_t^*.\text{candidate}\big) \\ & \wedge \big(\hat{y}_t.\text{work_before} = y_t^*.\text{work_before}\big) \\ & \wedge \big(\hat{y}_t.\text{work_after} = y_t^*.\text{work_after}\big) \\ & \wedge \big(\hat{y}_t.\text{finish} = y_t^*.\text{finish}\big) \end{aligned} \tag{2}$$

**Buddy System / Semaphore / Mutex:** Exact equality across all state fields (allocated block address-size tuples, free-list, counter/owner state, blocked thread list order).

*2) Diagnostic Trace Metrics:*

1) **Step Accuracy (SA):**

$$\text{SA} = \frac{1}{T^*} \sum_{t=1}^{\min(T^*, m)} \mathbb{I}(\hat{y}_t = y_t^*) \tag{3}$$

2) **First Error Position (FEP):**

$$\text{FEP} = \min\big(\{t \in \{1, \ldots, T^*\} \mid \hat{y}_t \neq y_t^*\} \cup \{\infty\}\big) \tag{4}$$

3) **Error Cascade Rate (ECR):**

$$\text{ECR} = \begin{cases} \dfrac{1}{T^* - \text{FEP}} \displaystyle\sum_{t=\text{FEP}+1}^{T^*} \mathbb{I}(\hat{y}_t \neq y_t^*), & \text{if FEP} < T^* \\ 0, & \text{if FEP} \geq T^* \text{ or FEP} = \end{cases} \tag{5}$$

4) **Verdict Accuracy (VA):** To prevent conflating parse failures with reasoning errors, we report both $\text{VA}_{\text{parsed}}$ (conditionally over successfully formatted outputs) and $\text{VA}_{\text{all}}$ (treating unparsed outputs as incorrect):

$$\text{VA}_{\text{parsed}} = \frac{1}{N_{\text{parsed}}} \sum_{i=1}^{N_{\text{parsed}}} \mathbb{I}(\widehat{\text{verdict}}_i = \text{verdict}_i^*) \tag{6}$$

$$\text{VA}_{\text{all}} = \frac{1}{N_{\text{total}}} \sum_{i=1}^{N_{\text{total}}} \mathbb{I}\Big(\text{parsed}_i \wedge \widehat{\text{verdict}}_i = \text{verdict}_i^*\Big) \tag{7}$$

TABLE I
SVAC-CONCURRENCY BENCHMARK SUITE OVERVIEW

| Algorithm Family | State Representation | Instances | Complexity Bounds | Class Balance |
|---|---|---|---|---|
| Wait-For Graph (WFG) | Directed Adjacency & Recursion Stack | 30 | $\|V\| \in [2,6]$, $\|E\| \in [1,8]$ | 50% Deadlock (15) / 50% Acyclic (15) |
| Banker's Algorithm | Available Vector, Allocation/Need Matrix | 30 | $P \in [2,6]$, $R \in [2,4]$ | 50% Safe (15) / 50% Unsafe (15) |
| Buddy System Memory | Free-list Binary Tree & Blocks | 25 | 512 KB–4096 KB, Requests $\in [4,10]$ | Deterministic Partition Trace |
| Counting Semaphore | `S.value` & FIFO Blocked Queue | 20 | $S \in [1,3]$, Ops $\in [4,10]$ | Deterministic State Sequence |
| Mutex Lock Synchronization | Owner ID & Lock State Flag | 20 | Threads $\in [2,5]$, Ops $\in [4,8]$ | Deterministic Transition Trace |
| **Total** | | **125** | 5 Complexity Tiers | 50.0% Majority Baseline |

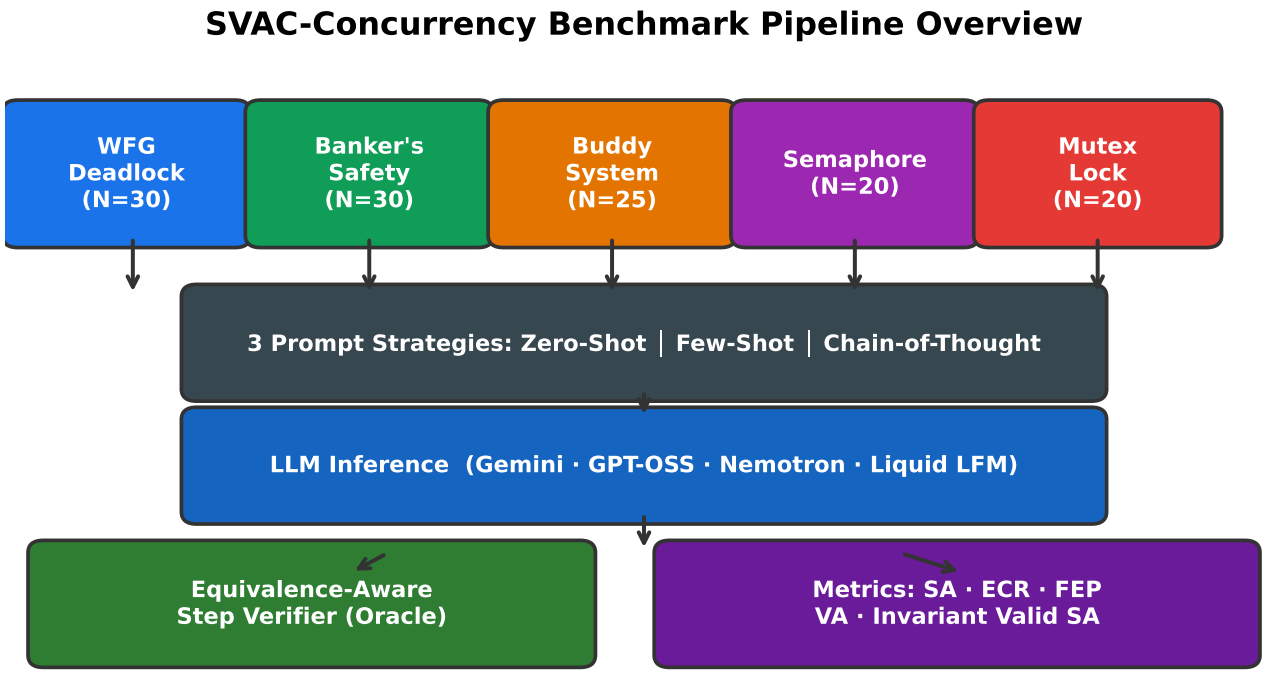



Fig. 1. SVAC-Concurrency benchmark pipeline: five algorithm families (WFG, Banker's, Buddy, Semaphore, Mutex) are evaluated across three prompting strategies. Each LLM response is scored by an equivalence-aware step verifier producing SA, ECR, FEP, VA, and Invariant Valid SA metrics.

*3) Equivalence-Aware Invariant Scoring:* Strict positional matching penalizes models when multiple valid execution paths exist (e.g., Banker's instances where multiple processes satisfy $\mathbf{Need}_i \leq$ Work, or graph traversals with initial visited set definition differences). We therefore formulate **Equivalence-Aware Invariant Validity ($\mathbf{SA_{valid}}$)**:

- **Banker's Safety:** At each step $t$, verify that candidate process $P_i$ is unfinished and satisfies $\mathbf{Need}_i \leq \text{Work}_{t-1}$, that $\text{Work}_t = \text{Work}_{t-1} + \mathbf{Alloc}_i$, and that if safe, the emitted sequence forms an executable permutation. (Note: selecting a different valid candidate than the reference solver's index-order scan represents an instruction deviation rather than solver ambiguity).
- **WFG Deadlock:** At each step $t$, verify that: label=iii)
  1) Edge validity: if action = visit, edge $(\text{stack}[-1], \text{node}) \in E$;
  2) Backtrack legality: if action = backtrack, node = $\text{stack}[-1]$;
  3) Stack consistency: recursion stack matches valid active search path;
  4) Final verdict: reported deadlock state matches ground-truth graph property.

## IV. EXPERIMENTAL SETUP

All evaluations were executed under strict deterministic parameters (temperature = 0.0, top_p = 1.0) across three prompt configurations: **Zero-Shot (ZS)** (direct JSON schema specification), **Few-Shot (FS)** (in-context exemplars of step-

by-step execution), and **Chain-of-Thought (CoT)** (verbalized rationale preceding structured JSON emission).

We evaluated four diverse foundation models:

1) **Google Gemini-3.5-Flash-Lite:** High-throughput dense instruction-tuned reasoning model evaluated via Google AI Studio API endpoints across all 125 instances ($N$=375).
2) **OpenAI GPT-OSS-20B:** Open-weight model evaluated via Groq high-speed LPU inference acceleration, scoped to WFG deadlock detection ($N$=81 scored attempts) due to upstream API rate-limits.
3) **NVIDIA Nemotron-3-Super-120B:** 120B-parameter open-weight foundation model evaluated via OpenRouter endpoints across all 125 instances ($N$=375 attempts).
4) **Liquid AI LFM-2.6B:** Non-transformer dynamical state-space architecture evaluated via OpenRouter endpoints, scoped to WFG ($N$=39 scored attempts).

## V. Results and Empirical Analysis

### A. The "Right Verdict for Wrong Reasons" Gap

A primary finding from our benchmark is the stark divergence between Verdict Accuracy and Step Accuracy. As shown in Table II, Gemini-3.5-Flash-Lite achieves an overall **Verdict Accuracy of 72.8%** on parsed responses, yet exhibits a mean **Step Accuracy of only 56.1%**.

Crucially, the 56.1% suite mean is heavily inflated by trivial state saturation in Mutex (100.0% SA) and Semaphore tasks (94.2% SA). When isolating the **non-saturating algorithmic tasks (WFG, Banker's, and Buddy System)**, mean Step Accuracy drops to **36.98%**. Because the decision tasks are strictly 50/50 balanced (majority baseline $= 50.0\%$), the gap between 72.8% VA and 37.0% non-saturated SA demonstrates that outcome accuracy vastly overstates procedural competence.

### B. Deconstructing the CoT Inversion Effect: Priming vs. Invariants

Table III appears to show an extreme prompting anomaly. In linear-procedural tasks, CoT prompting provides substantial gains ($+16.8\%$ Banker's, $+18.6\%$ Buddy over Zero-Shot). In WFG cycle detection, strict positional matching reports an apparent collapse: Few-Shot attains **85.4% SA** whereas CoT plunges to **7.5% SA**.

A superficial reading concludes that CoT verbalization causes general topological reasoning breakdown. However, cross-examining Table IV and Table V reveals a deeper, nuanced scientific truth:

1) **High Invariant Validity and Verdict Accuracy:** Under invariant checking, Zero-Shot achieves **93.7% valid SA** (90.0% VA), Few-Shot achieves **100.0% valid SA** (100.0% VA), and CoT achieves **93.2% valid SA** (93.3% VA). Models across all three strategies execute sound DFS traversals.
2) **Why Strict SA Collapses (Format Priming & Set Conventions):** In Few-Shot prompting, the in-context exemplar explicitly demonstrates the solver's convention of adding nodes to `visited_set` on initial entry (`["P0"]`). Without this in-context exemplar, Zero-Shot and CoT models strictly follow the prompt text definition (`visited_set` = nodes fully explored/backtracked so far), returning `[]` on step 1. Because strict positional matching requires exact field equality at step $t$, this initial set convention mismatch causes step 1 failure in 100% of ZS and CoT cases, artificially depressing strict positional SA to 4.4%–7.5% despite 93.2%–93.7% valid invariant execution.
3) **Residual CoT Narration Dynamics:** Under CoT prompting, verbalized step-by-step rationale shows a trend toward lower recursion stack corruption (8/30 in CoT vs. 14/30 in Zero-Shot) and fewer false cycles (2/30 in CoT vs. 3/30 in Zero-Shot), though these differences are not statistically significant at $N = 30$ (Fisher's exact test $p = 0.18$ and $p = 1.00$). However, `visited_set_corruption` remains 100% under strict positional matching due to the prompt/solver convention mismatch described above.

**Case Study: Localized CoT Backtrack Drift (Instance `wfg_3ceec919261d`).** Ground truth for this instance is `deadlock=False` (DAG). At step 9, the model's trace correctly backtracks P4 out of the recursion stack (`"recursion_stack": ["P0"]`), marking P4 as fully explored. However, 12 steps later at step 21, the model re-asserts P4 as being "in the recursion stack" and declares a false cycle—contradicting its own step-9 state. This demonstrates how unconstrained natural language tokens can induce localized state-binding drift at terminal steps.

### C. First Error Position (FEP) Dynamics

Evaluating the discrete point of first failure (FEP) reveals fundamentally distinct degradation dynamics across algorithmic domains:

- **Immediate Convention Divergence (WFG):** Under strict positional matching, FEP $= 1$ in 100.0% of error cases for ZS and CoT (mean FEP $= 1.00$) due to the initial node `visited_set` entry convention mismatch. Under Few-Shot, format priming aligns initial set conventions, delaying initial failure to mean FEP $= 8.38$ (median 7.0).
- **Early Matrix Misalignment (Banker's):** FEP $= 1$ occurs in 76.0% (ZS), 59.1% (FS), and 82.4% (CoT) of error instances (mean FEP $\in [1.28, 1.55]$).
- **Gradual Multi-Step State Drift (Buddy System):** FEP $= 1$ occurs in only 22.7%–34.8% of errors, with a mean FEP of **6.96 steps (ZS), 7.57 (FS), and 6.55 (CoT)** (median FEP $= 4.0$–$5.0$). Models correctly perform initial block splits, but accumulate arithmetic fragmentation errors over long horizons.

### D. Equivalence-Aware Invariant Scoring vs. Strict Positional Matching

As detailed in Table V:

TABLE II
MULTI-PROVIDER CROSS-ARCHITECTURE BENCHMARK RESULTS

| Model Architecture | Organization | Att. | Parse | SA | ECR | $VA_{parsed}$ | $VA_{all}$ |
|---|---|---|---|---|---|---|---|
| Gemini-3.5-Flash-Lite (Full Suite) | Google DeepMind | 375 | 99.2% | 56.1% | 36.9% | 72.8% | 72.3% |
| Gemini-3.5-Flash-Lite (WFG Only) | Google DeepMind | 90 | 97.8% | 31.2% | 73.8% | 94.3% | 92.2% |
| GPT-OSS-20B (WFG Only) | OpenAI | 90† | 58.9% | 53.1%* | 45.3%* | 100.0%* | 58.9% |
| Nemotron-3-Super-120B (Full Suite) | NVIDIA | 375 | 22.1% | 52.9%* | 38.0%* | 96.4%* | 21.3% |
| Liquid LFM-2.6B (WFG Only) | Liquid AI | 90† | 30.0% | 14.4%* | 86.1%* | 96.3%* | 30.0% |
| Binary Majority Baseline‡ | — | 375 | 100.0% | — | — | 50.0% | 50.0% |

*Computed conditionally over successfully parsed JSON outputs (Nemotron $N_{parsed} = 83$, GPT-OSS $N_{parsed} = 53$, LFM $N_{parsed} = 27$).
†Scoped to WFG subset due to free-tier upstream rate-limiting (HTTP 429) and network connection resets.
‡Majority baseline of 50.0% applies strictly to binary decision tasks (WFG deadlock, Banker's safety).
Parse Rate $= N_{parsed}/N_{attempted}$. Att. = total requested attempts.

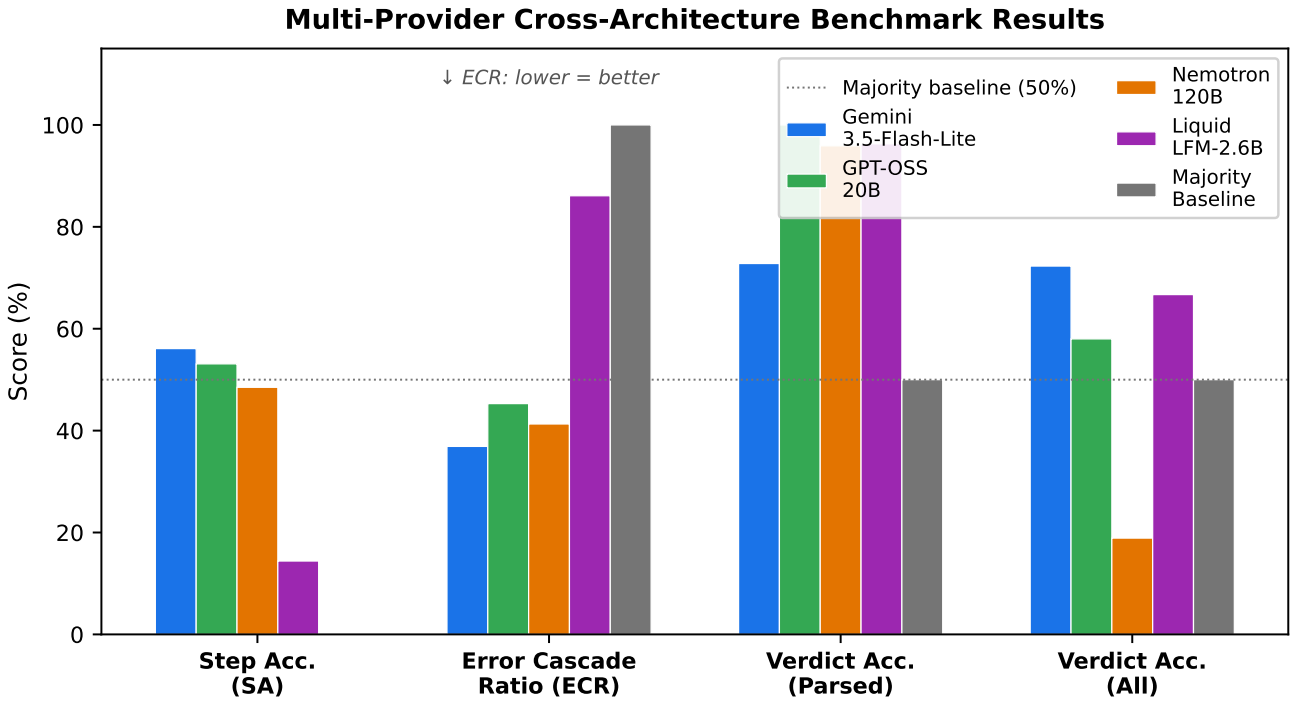



Fig. 2. Cross-architecture benchmark results. SA and VA (All) are the primary metrics; ECR (lower is better) quantifies error propagation. Nemotron's low VA (All) of 18.9% reflects its 19.7% parse rate. Liquid LFM-2.6B's near-zero SA despite high VA reveals the Right-Verdict-for-Wrong-Reasons failure mode.

1) **Banker's Algorithm:** When verifying that each chosen candidate process satisfies $\mathbf{Need}_i \leq$ Work and properly reclaims resources under the model's own state, valid step accuracy reaches **74.3%** (+32.8% over positional SA). In 7 out of 90 instances, models executed a sound safe sequence that prioritized a different candidate process than the reference solver's deterministic tie-break.
2) **WFG Cycle Detection:** Under invariant checking (valid neighbor traversal and backtracking), WFG valid step accuracy reaches **95.6%** (+64.3% over positional SA).
3) **Trace Length Alignment Sensitivity:** Evaluating monotonic Longest Common Subsequence (LCS) alignment reveals that trace length disparity ($m/T^* \approx$ 1.3–2.8) creates substantial artificial cascade penalties: aligned SA increases Banker's CoT from 49.7% to 63.5% (+13.8%) and WFG Few-Shot from 85.4% to 97.7% (+12.3%).

### E. Cross-Model Schema Fragility and Formatting Brittleness

Table II underscores a profound divergence between proprietary instruction-tuned models and open-weight architectures. While Gemini-3.5-Flash-Lite achieved 99.2% parse success, **NVIDIA Nemotron-3-Super-120B achieved only 19.7% parse success**, with the remaining 80.3% of outputs failing JSON decoding due to missing closing delimiters, extraneous conversational preambles, or unescaped tokens.

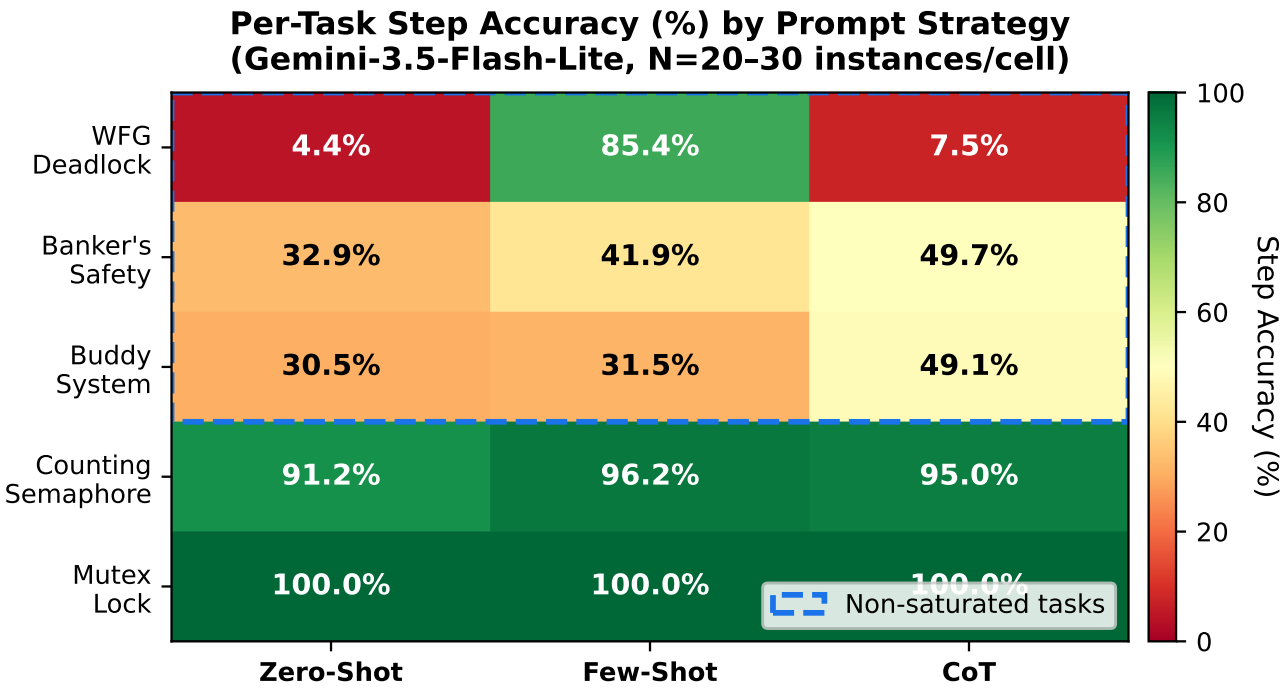



Fig. 3. Step Accuracy heatmap across five task domains and three prompt strategies (Gemini-3.5-Flash-Lite). The dashed blue rectangle highlights non-saturating algorithmic tasks (WFG, Banker's, Buddy System) where mean SA drops to 36.98%, exposing the fundamental limits of LLM procedural reasoning.

TABLE III
PER-TASK STEP ACCURACY (%) ACROSS PROMPT STRATEGIES
(GEMINI-3.5-FLASH-LITE, $N$ = 20–30 INSTANCES PER CELL)

| **Task Domain** | **ZS** | **FS** | **CoT** |
|---|---|---|---|
| WFG Deadlock | 4.4% | 85.4% | 7.5% |
| Banker's Safety | 32.9% | 41.9% | 49.7% |
| Buddy System | 30.5% | 31.5% | 49.1% |
| Counting Semaphore | 91.2% | 96.2% | 95.0% |
| Mutex Lock | 100.0% | 100.0% | 100.0% |
| Non-Saturated Mean‡ | 22.6% | 52.9% | 35.4% |
| Instance-Weighted Mean♮ | 45.6% | 68.2% | 54.7% |

‡WFG + Banker's + Buddy only. Overall non-saturated SA: 36.98%.
♮Weighted by $N$ per task (30,30,25,20,20); strategy mean = 56.2%, reconciling to suite SA of 56.1%.

TABLE IV
WFG EMPIRICAL ERROR TAXONOMY & ACCURACY BREAKDOWN
(GEMINI)

| **Metric / Error Category** | **ZS** ($N$=30) | **FS** ($N$=30) | **CoT** ($N$=30) |
|---|---|---|---|
| Visited Set Corruption | 30 (100%) | 7 (23.3%) | 30 (100%) |
| Recursion Stack Corr. | 14 (46.7%) | 12 (40.0%) | 8 (26.7%) |
| Wrong Traversal Node | 20 (66.7%) | 12 (40.0%) | 18 (60.0%) |
| Step Count Mismatch | 16 (53.3%) | 13 (43.3%) | 12 (40.0%) |
| False Cycle Hallucination | 3 (10.0%) | 0 (0.0%) | 2 (6.7%) |
| Missed Cycle (False Neg.) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) |
| Verdict Acc. (VA) | 27/30 (90.0%) | 28/28 (100%)§ | 28/30 (93.3%) |
| Strict Positional SA | 4.4% | 85.4% | 7.5% |
| Invariant-Based Valid SA | 93.7% | 100.0% | 93.2% |

§2 Few-Shot instances failed schema parsing; VA computed over 28 parsed responses.

Crucially, auditing the parsed subset reveals that Nemotron achieved **95.9% Verdict Accuracy** and **48.5% Step Accuracy**. Our dual-metric decomposition ($VA_{parsed}$ vs. $VA_{all}$) establishes that open-weight models suffer not from semantic reasoning deficits, but from strict output-channel schema adherence failures.

## VI. DISCUSSION AND PRACTICAL IMPLICATIONS

### A. Token-Level Format Following vs. Algorithmic Competence

Our findings highlight that raw parameter count does not ensure procedural fidelity. Nemotron-120B contains 120B total parameters, yet failed schema parsing on 80.3% of tasks where Gemini-3.5-Flash-Lite parsed cleanly. For practical systems verification pipelines, constrained decoding (e.g., grammar-based sampling via outlines or JSON schemas) is essential before open-weight models can be reliably deployed.

### B. Implications for Automated Verification and Systems Pedagogy

1) **Never rely on unverified verdicts:** On balanced 50/50 concurrency distributions, LLMs exhibit a 35.8% divergence between outcome correctness and procedural fidelity. Downstream systems tools must verify intermediate execution traces.
2) **Distinguish between format compliance and procedural validity:** In graph and state-machine problems,

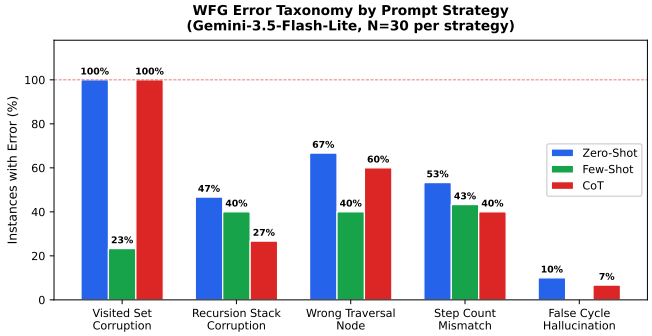



Fig. 4. WFG error taxonomy by prompt strategy. Few-Shot dramatically reduces Visited Set Corruption (100%→23.3%) via format priming. False Cycle Hallucination is absent in Few-Shot but rises in CoT, indicating localized state-binding drift under verbalized traces.

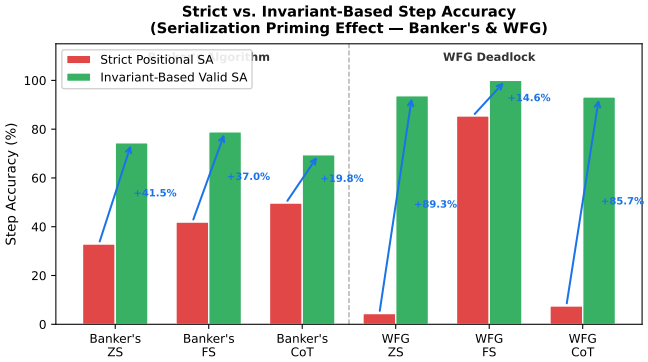



Fig. 5. Strict vs. invariant-based step accuracy. Blue arrows show the equivalence gap $\Delta$. WFG Zero-Shot improves from 4.4% to 93.7% (+89.4%) under invariant scoring, demonstrating that models reason correctly but deviate from the canonical traversal order imposed by strict positional matching.

TABLE V
INVARIANT-BASED VALID SA VS. STRICT POSITIONAL SA (GEMINI)

| Task / Variant | Strict SA | Valid SA | $\Delta$ |
|---|---|---|---|
| *Banker's Safety ($N$=90)* | | | |
| Zero-Shot | 32.9% | 74.4% | +41.6 pp |
| Few-Shot | 41.9% | 78.9% | +37.1 pp |
| CoT | 49.7% | 69.5% | +19.8 pp |
| **Mean** | **41.5%** | **74.3%** | **+32.8** pp |
| *WFG Deadlock ($N$=88)* | | | |
| Zero-Shot | 4.4% | 93.7% | +89.4 pp |
| Few-Shot | 85.4% | 100.0% | +14.6 pp |
| CoT | 7.5% | 93.2% | +85.8 pp |
| **Mean** | **31.2%** | **95.6%** | **+64.3** pp |

few-shot exemplars serve primarily to prime arbitrary serialization conventions (tie-breaks). Automated benchmarks must adopt equivalence-aware verifiers to avoid mistaking valid alternative traversals for reasoning failures.

3) **Constrain generative freedom in topological tasks:** Unstructured verbalization in CoT increases state-binding drift on recursion stacks. Structured schema emission should be enforced.

## VII. LIMITATIONS

This empirical investigation is bounded by several methodology considerations:

1) **Single-Run Deterministic Sampling:** Inferences were performed at temperature $= 0.0$. We did not evaluate temperature-swept self-consistency or majority voting across multiple runs.
2) **Per-Cell Sample Size:** With 125 instances spanning 5 tasks and 5 complexity levels, individual task-complexity sub-cells contain $N = 4$–$6$ instances, though each prompt strategy evaluates $N = 20$–$30$ instances per algorithm family.
3) **Partial Evaluation of Open Models:** Upstream API rate-limiting on open-router endpoints restricted open-weight evaluations (GPT-OSS-20B and LFM-2.6B) to WFG subsets, necessitating conditional reporting over parsed subsets.
4) **Synthetic Problem Horizons:** Instances model canonical textbook concurrency algorithms ($P \leq 6, R \leq 4$) rather than production-scale distributed systems traces.
5) **Static Prompting Regimes:** We evaluated static ZS, FS, and CoT prompts without dynamic tool feedback, external Python interpreter execution, or self-correction loops.
6) **Tie-Breaking Sensitivity:** While invariant-aware scoring resolves alternative valid sequences, strict positional SA remains sensitive to canonical solver ordering.

## VIII. FUTURE WORK

Future extensions of SVAC-Concurrency will investigate:

1) **Grammar-Constrained Decoding:** Evaluating whether formal context-free grammars (CFGs) eliminate open-weight formatting collapse.
2) **Reinforcement Learning with Step Verifiers:** Utilizing SVAC's intermediate step verifiers as dense process reward models (PRMs) to train reasoning models on discrete concurrency invariants.
3) **Kernel-Level Concurrency Traces:** Extending problem instances to real Linux kernel eBPF trace logs and lockdep synchronization graphs.

## IX. CONCLUSION

We presented **SVAC-Concurrency**, a fine-grained, open-source benchmark for evaluating LLM intermediate reasoning traces on Operating System concurrency algorithms. Through multi-dimensional metrics (SA, FEP, ECR, and Equivalence-Aware Validity), we showed that high outcome accuracy masks severe intermediate state failures, deconstructed the CoT Inversion Effect in topological deadlock detection as serialization tie-breaking priming, and demonstrated the critical impact of invariant-aware scoring. All benchmark datasets, reference solvers, and evaluation harnesses are made publicly available to advance verifiable algorithmic reasoning.

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